9/16/2026

BASICS OF HTML

Full-Stack Development Lab

Instructor: Sir M. Fahad Qasim

Submitted by: Wasif Ali

Reg id: 245187

Class: BSCS F24 B



Lab No 01

Task 01

Create an interactive Stack Visualizer using HTML, CSS, JavaScript, and jQuery. Click a value to push it into the stack and click a value already in the stack to pop it. Handle invalid POP operations appropriately.

Code:

Index.html

```
<!DOCTYPE html>
<html lang="en">

<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Stack Visualizer</title>
    <link rel="stylesheet" href="style.css">
    <!-- Include jQuery -->
    <script src="https://code.jquery.com/jquery-3.6.0.min.js"></script>
</head>

<body>
    <h2>Stack Visualizer (LIFO)</h2>
    <p>Click a number to PUSH it to the stack:</p>

    <div class="input-values">
        <button class="push-btn">10</button>
        <button class="push-btn">20</button>
        <button class="push-btn">30</button>
        <button class="push-btn">40</button>
    </div>

    <p>Click the top item in the stack to POP it:</p>
    <div id="stack-container">

    </div>

    <script src="script.js"></script>
</body>

</html>
```

Style.css

```css
body {
    font-family: Arial, sans-serif;
    text-align: center;
    margin: 40px;
}

.input-values button {
    padding: 10px 20px;
    margin: 5px;
    font-size: 16px;
    cursor: pointer;
    background-color: #4CAF50;
    color: white;
    border: none;
    border-radius: 4px;
}

#stack-container {
    width: 200px;
    min-height: 250px;
    margin: 30px auto;
    border: 4px solid #333;
    border-top: none;
    display: flex;
    flex-direction: column;
    justify-content: flex-end;
    padding: 10px;
    background-color: #f9f9f9;
}
.stack-item {
    background-color: #2196F3;
    color: white;
    padding: 15px;
    margin-bottom: 5px;
    text-align: center;
    font-weight: bold;
    cursor: pointer;
    border: 1px solid #1976D2;
}
.stack-item:hover {
    background-color: #1976D2;
}
```

Script.js

```
$(document).ready(function () {


    $('.push-btn').click(function () {

        let value = $(this).text();


        let newItem = $('<div class="stack-item"></div>').text(value);


        $('#stack-container').prepend(newItem);
    });

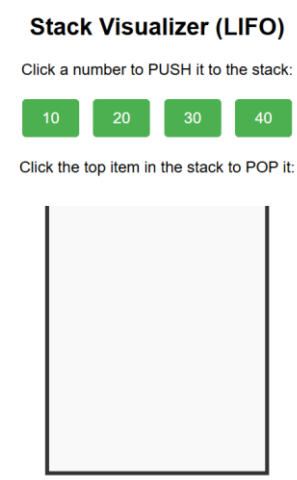

    $('#stack-container').on('click', '.stack-item', function () {

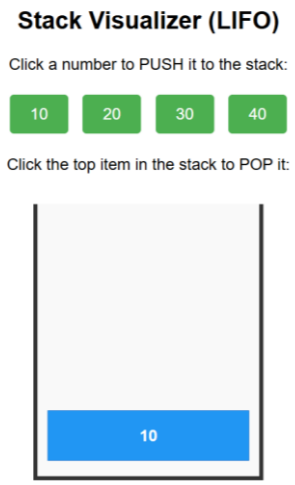

        if ($(this).is(':first-child')) {
            $(this).remove();
        } else {

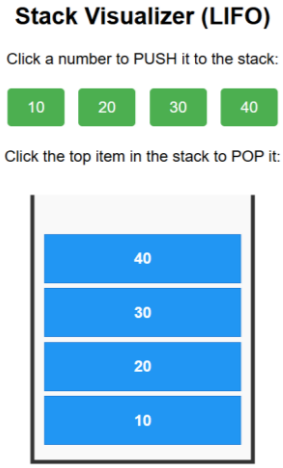
            alert("Invalid POP! In a Stack, you can only pop the top-most
element.");
        }
    });

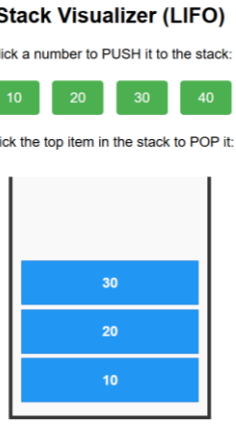
});
```

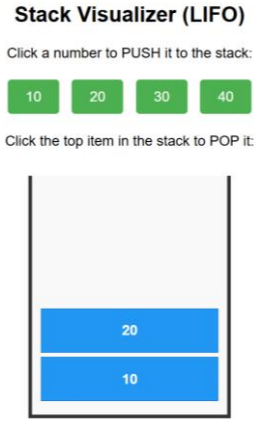
OUTPUT:

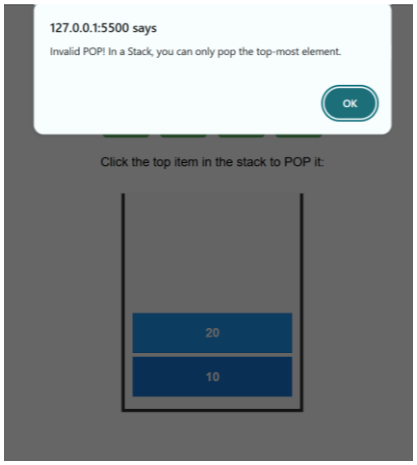

Stack Visualizer (LIFO)
Click a number to PUSH it to the stack:
10
20
30
40
Click the top item in the stack to POP it:


Stack Visualizer (LIFO)
Click a number to PUSH it to the stack:
10
20
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Stack Visualizer (LIFO)
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Click a number to PUSH it to the stack:
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Click the top item in the stack to POP it:
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10


Stack Visualizer (LIFO)
Click a number to PUSH it to the stack:
10
20
30
40
Click the top item in the stack to POP it:
20
10


127.0.0.1:5500 says
Invalid POP! In a Stack, you can only pop the top-most element.
OK
Click the top item in the stack to POP it:
20
10

Task 02

Create an interactive Queue Visualizer using HTML, CSS, JavaScript, and jQuery. Click a value to enqueue it and click a value already in the queue to dequeue it. Handle invalid DEQUEUE operations appropriately.

Code:

Index.html

```
<!DOCTYPE html>
<html lang="en">

<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Queue Visualizer</title>
    <link rel="stylesheet" href="style.css">
    <!-- Include jQuery -->
    <script src="https://code.jquery.com/jquery-3.6.0.min.js"></script>
</head>

<body>
    <h2>Queue Visualizer (FIFO)</h2>
    <p>Click a number to ENQUEUE (add to back):</p>

    <div class="input-values">
        <button class="enq-btn">A</button>
        <button class="enq-btn">B</button>
        <button class="enq-btn">C</button>
        <button class="enq-btn">D</button>
    </div>

    <p>Click the front item in the queue to DEQUEUE it:</p>
    <div class="queue-wrapper">
        <span class="label">Front (Exit) &lt;--</span>
        <div id="queue-container">

        </div>
        <span class="label">&lt;-- Back (Entry)</span>
    </div>
    <script src="script.js"></script>
</body>
</html>
```

Style.css

```css
body {
    font-family: Arial, sans-serif;
    text-align: center;
    margin: 40px;
}

.input-values button {
    padding: 10px 20px;
    margin: 5px;
    font-size: 16px;
    cursor: pointer;
    background-color: #FF9800;
    color: white;
    border: none;
    border-radius: 4px;
}

.queue-wrapper {
    display: flex;
    align-items: center;
    justify-content: center;
    margin-top: 30px;
}

.label {
    font-weight: bold;
    color: #555;
    margin: 0 15px;
}

#queue-container {
    display: flex;
    flex-direction: row;
    /* Horizontal line */
    min-width: 300px;
    height: 60px;
    border-top: 4px solid #333;
    border-bottom: 4px solid #333;
    background-color: #f9f9f9;
    padding: 5px;
    overflow-x: auto;
}
```

```css
.queue-item {
    background-color: #9C27B0;
    color: white;
    min-width: 50px;
    height: 100%;
    margin-right: 5px;
    display: flex;
    align-items: center;
    justify-content: center;
    font-weight: bold;
    cursor: pointer;
    border: 1px solid #7B1FA2;
}

.queue-item:hover {
    background-color: #7B1FA2;
}
```

Script.js

```javascript
$(document).ready(function () {

    $('.enq-btn').click(function () {

        let value = $(this).text();


        let newItem = $('<div class="queue-item"></div>').text(value);

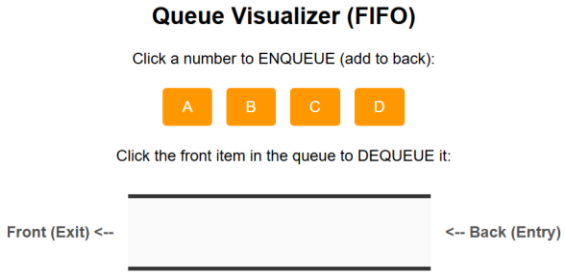

        $('#queue-container').append(newItem);
    });

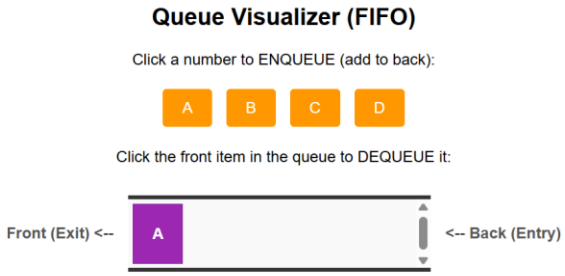

    $('#queue-container').on('click', '.queue-item', function () {

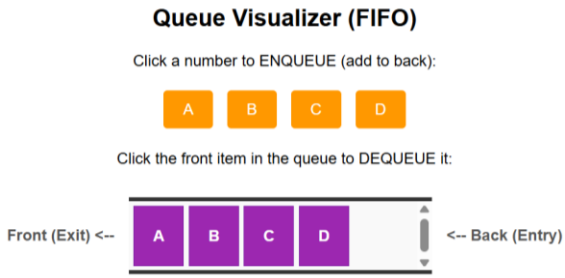
        if ($(this).is(':first-child')) {
            $(this).remove();
        } else {
            alert("Invalid DEQUEUE! In a Queue, you can only remove the element
at the front.");
        }
    });

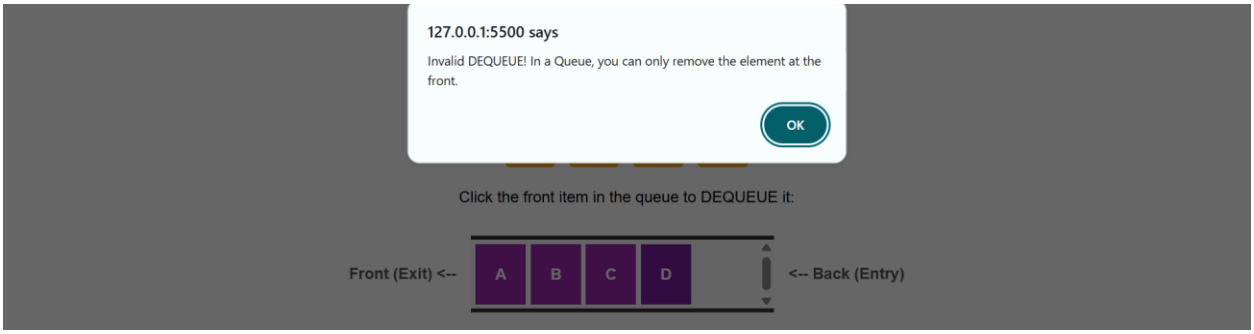
});
```

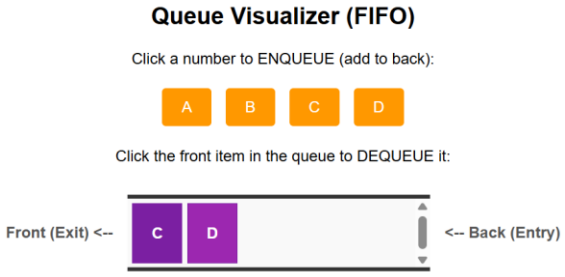
OUTPUT:


Queue Visualizer (FIFO)
Click a number to ENQUEUE (add to back):
A
B
C
D
Click the front item in the queue to DEQUEUE it:
Front (Exit) <--
<-- Back (Entry)


Queue Visualizer (FIFO)
Click a number to ENQUEUE (add to back):
A
B
C
D
Click the front item in the queue to DEQUEUE it:
Front (Exit) <--
A
<-- Back (Entry)


Queue Visualizer (FIFO)
Click a number to ENQUEUE (add to back):
A
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D
Click the front item in the queue to DEQUEUE it:
Front (Exit) <--
A
B
C
D
<-- Back (Entry)


127.0.0.1:5500 says
Invalid DEQUEUE! In a Queue, you can only remove the element at the front.
OK
Click the front item in the queue to DEQUEUE it:
Front (Exit) <--
A
B
C
D
<-- Back (Entry)


Queue Visualizer (FIFO)
Click a number to ENQUEUE (add to back):
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Click the front item in the queue to DEQUEUE it:
Front (Exit) <--
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<-- Back (Entry)